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Multiphysics modeling and simulation of selective laser sintering manufacturing processes

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Abstract

A significant percentage of materials used in industry start in particulate form. In many modern applications, these systems undergo processing which necessitate a multiphysical analysis. Several manufacturing applications have arisen that involve the multiphysical response of particulate systems in the presence of strongly coupled electromagnetic, optical, and thermal fields. The significant multifield coupling requires methods that can capture the unique and essential physics of these systems. Specifically, in this work, the modeling and simulation of selective laser sintering of particulate materials is discussed. Such processes involve harnessing optical energy to heat and fuse powdered materials together in an additive process. Selective laser sintering allows for the rapid manufacturing or prototyping of parts with complex geometries. In order to simulate such a process in a rapid manner, the approach pursued by the authors is to develop a computational tool by assembling relatively simple, physically meaningful, models at the small scale, for many interacting particles. This allows for much more refined estimates of the resulting overall system temperature and, ultimately, its change of phase from a solid, to a liquid, and possibly even to a gas. Large-scale three-dimensional examples are provided.

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1. Introduction

Selective laser sintering (SLS) is a type of rapid prototyping, tooling, and manufacturing (RP, RT, and RM) technique wherein a three dimensional object is built layer-by-layer out of a powdered material directly from a CAD model. The original process was invented in 1986 at the University of Texas at Austin [1]. Since then its use has increasingly grown for a wide variety of applications. Laser sintering has the advantage of being able to rapidly produce parts of relatively complex geometries, in a timely and cost effective manner. It is especially useful in its ability to make metallic parts, which would otherwise be impossible to form using other RP, RT, and RM techniques as shown in the works by Simchi and Pohl [2, 3]. Maeda and Childs and Fischer et al. [4, 5] have demonstrated the ability for laser sintering to produce parts made from hard metal powders,

such as WC-Co and TiC-Ni used in machining tools and abrasion resistant coatings. Laser sintering has also seen uses in RP and RM for polymeric and ceramic materials. Tan et al., Williams et al., and Schmidt et al. [6, 7, 8] have successfully laser sintered polymeric biomaterials which can be used to produce tissue engineering scaffolds and certain types of implants.

The process works by using localized heat from a laser to selectively fuse the powdered material, often via full melting of the particles or melting of a binder applied to the powdered particles. Typically either a continuous wave or modulated CO₂ laser is used with power ranging from 1 – 250 W. However, more recently Nd:YAG lasers have also been successfully used [9]. Rollers are used to evenly distribute a thin layer of the powders to be sintered upon a fabrication bed. Each layer is typically around the order of 100 microns thick. The laser is then directed (via a scanner system) over the powders in the pattern of the desired shape. The laser

beam typically has a nominal diameter on the order of 0.5 mm or less. The scan rate can vary between 2 and 200 cm/s. Once one layer of powder is sintered, the fabrication bed lowers and a new layer of powder is placed on top by the roller. This layer-by-layer, additive manufacturing process is repeated until the final part has been fabricated. Figure 1 illustrates the set-up of a typical SLS machine.

To aid in the sintering process, frequently the sintering chamber is heated to minimize required laser energy and expedite the process. Additionally, an inert gas, such as nitrogen, is often present to avoid oxidation or burning of the powders. Upon completion of the sintering process, the final part is removed and often bead blasted to remove unsintered, adhering particles [1]. To increase the density of the final part, additional operations, such as hot isostatic pressing or infiltration, are also required depending on the application [9].

The aim of this work is to produce a computational model of the SLS process whereby process parameters (laser power, scan rate, etc.) can be optimized for the sintering of a given part and material. Such a model can serve as a useful tool for operators and greatly reduce the need for costly experiments. Many researchers have already proposed different ways to simulate this process. Kolossov et al. and Dong et al. [10, 11] have created 3D finite element (FE) models for the temperature evolution during laser sintering. The latter model also predicts densification. Matsumoto et al. [12] proposed a FE method for calculating the temperature and stress distribution in a single layer of sintered material. Nelson et al. [13] used empirical data to create a 1D heat transfer model of SLS that can predict sintering depths. Simchi and Simchi and Pohl [2, 3] used empirical results to determine a relationship between energy input and densification during SLS.

In the previously described works, the material was treated as a continuum medium and effective material properties were used. Additionally, due to the difficulty of accounting for localized phase change in FE models, this effect was usually neglected. Empirical models have the drawback of often being process and material specific. In this work a discrete element model is proposed to model the deposition and subsequent laser sintering of the powdered particles. An algorithm for dealing with the change in material properties due to phase changes is also presented. 3D simulation results are shown and discussed. Finally conclusions are drawn, along with future work to be done on this topic.

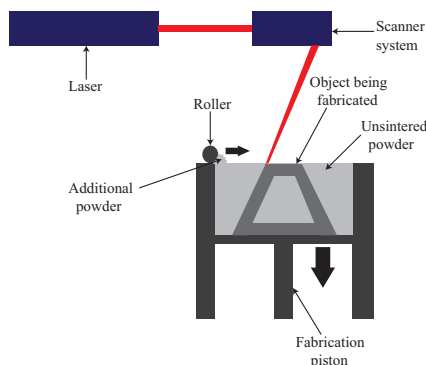


Fig. 1. Schematic of a typical SLS set-up

2. Modeling Approach

A multiphysical modeling approach has been employed to simulate the selective laser sintering process for a single layer of particles. A discrete element approach was used where particle-to-particle and particle-to-wall mechanical and thermal interactions are considered. Each particle is idealized as a sphere and the temperature of each sphere is assumed to be uniform throughout that individual particle. The modeling approach can be characterized in two parts: (1) simulation of the deposition of the powder particles; (2) simulation of the temperature evolution as a laser beam is passed over the particles. The model follows approaches previously developed in other works by Zohdi [14, 15].

The thickness of the simulated layer is approximately 100 μm . A truncated Gaussian distribution of particle sizes was used. The mean particle diameter is 50 μm with a standard deviation of 20 μm . The particle diameter is constrained to be between 20 μm and 100 μm . A small representative domain of cross-section 2 mm x 2 mm is used to reduce computation time while simultaneously being large enough to show the aggregate effects of particle-to-particle interactions.

2.1. Particle Dynamics

A simple model of the deposition of the particles is described in this section. In an effort to simplify the layer deposition process, the authors assume that the particles are being dropped into the above-mentioned box from a height ranging from 0.6 to 0.9 mm.

Starting from Newton's second law, we can determine an equation for the motion of the i -th particle (starting from a sample of N non-intersecting particles):

$$m_i \ddot{\mathbf{r}}_i = \Psi_i^{total} = \Psi_i^{con} + \Psi_i^{fric} + \Psi_i^{adh} + \Psi_i^{env} + \Psi_i^{grav} \quad (1)$$

where m represents mass, $\mathbf{r}$ represents position, and Ψ represents force.

The contact force, Ψ_i^{con} , is modeled via an overlap model where if the distance between two particles (i and j) is less than the combined radii of the particles then a particle-to-particle contact force exists [14]:

If $\|\mathbf{r}_i - \mathbf{r}_j\| \leq R_i + R_j$, then

$$\Psi_{ij}^{con} = \beta \epsilon_{ij} \mathbf{n}_{ij} \quad (2)$$

where ϵ_{ij} represents the amount of overlap given by

$$\epsilon_{ij} = \frac{\|\mathbf{r}_i - \mathbf{r}_j\| - (R_i + R_j)}{(R_i + R_j)} \quad (3)$$

and $\mathbf{n}_{ij}$ is the unit normal vector between the particle centers

$$\mathbf{n}_{ij} = \frac{\mathbf{r}_j - \mathbf{r}_i}{\|\mathbf{r}_j - \mathbf{r}_i\|} \quad (4)$$

Note that in the above equations R represents the radii of the particles and β is a penalty parameter to minimize particle

overlap. The contact force between particles and the walls is modeled in exactly the same manner except that the position of particle j is replaced by the position of the wall. The value of the penalty parameter for particle-to-wall collisions is greater than that for particle-to-particle collisions to model the more rigid walls. In other words, $\beta_{wall} > \beta_{particle}$ in Equation 2. The contact force on a particle is equal to the vector sum of all the contact forces acting upon it from other particles and/or walls.

For the forces due to friction, Ψ_i^{fric} , we assume a continuous sliding model. That is

$$\Psi_{ij}^{fric} = \mu_d \|\Psi_i^{con}\| \mathbf{t}_{ij} \quad (5)$$

where $\mathbf{t}_{ij}$ is the unit tangential vector between the particles, given by

$$\mathbf{t}_{ij} = \frac{\mathbf{v}_{tj} - \mathbf{v}_{ti}}{\|\mathbf{v}_{ti} - \mathbf{v}_{tj}\|} \quad (6)$$

In the above equation, $\mathbf{v}_t$ represents the tangential velocity, $\mathbf{v}_t = \mathbf{v} - (\mathbf{v} \cdot \mathbf{n})\mathbf{n}$. The friction between particles and the walls is modeled in the same manner except that the velocity of the wall (equal to $\mathbf{0}$ in this case) replaces the velocity of particle j . Figure 2 shows a pictorial representation of the contact and friction forces.

For the adhesion force, Ψ_i^{adh} between particles, we employ a criterion where if the particle overlap and average temperature between two particles are greater than set values, then a binding force is activated between the particles of the form:

$$\text{If } |\epsilon_{ij}| \geq \epsilon^* \text{ and } \frac{(\theta_i + \theta_j)}{2} > \theta^*, \text{ then}$$

$$\Psi_{ij}^{adh} = \gamma |\epsilon_{ij}| \mathbf{n}_{ij} \quad (7)$$

where γ is a constant.

The environmental force, Ψ_i^{env} , comes into play as a source of damping due to the surrounding environment (i.e. air interaction with the particles). Environmental forces additionally exist due to the presence of lubricant binders often placed in the powdered material (the green compact). A simple model for these environmental forces is

$$\Psi_i^{env} = -c^{env}(\mathbf{v}_i - \mathbf{v}^{env}) \quad (8)$$

where c^{env} is the environmental damping factor and $\mathbf{v}^{env}$ is the local velocity of the ambient medium (equal to $\mathbf{0}$ in the present work).

Finally there exists the gravitational force, Ψ_i^{grav} , acting along the $-\hat{\mathbf{z}}$ direction, given by

$$\Psi_i^{grav} = -m_i g \hat{\mathbf{z}}. \quad (9)$$

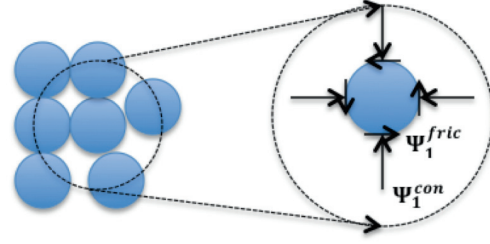


Fig. 2. Contact and friction forces on a particle

2.2. Thermal Effects – Heat Transfer

Assuming a lumped capacitance model where the temperature field is uniform throughout each particle (consistent with the low Biot numbers resulting from the small particle sizes), the thermal governing equation in integral form is:

$$\int_{\partial\omega} \mathbf{Q} \cdot \mathbf{n} dA + \int_{\omega} H dV = \int_{\omega} \rho C \frac{d\theta}{dt} \quad (10)$$

where $\mathbf{Q}$ represents heat transfer, H is a source term representing heat input from the laser, ρ is density, C is the heat capacity of the material, θ is temperature, and ω represents the domain of interest.

With the given particle sizes and the time scales present in this problem, it can be shown that the heat transfer through convection and radiation are more than one order of magnitude smaller than the heat transfer due to conduction. Performing a quick analytical analysis of the different heat transfer mechanisms in this problem, one can show that

$$Q_{conv}/Q_{cond} = \vartheta(10^{-2}) \text{ and } Q_{rad}/Q_{cond} = \vartheta(10^{-3}).$$

Note that the above ratio for convection is calculated using the highest possible heat transfer coefficient expected to be seen only during boiling or condensation. For normal free convection conditions, the ratio between convection and conduction would be closer to 10^{-4} . Thus, assuming that only conductive heat transfer occurs (see Fig. 3), we have

$$\sum_{j=1}^N K \frac{\theta_j - \theta_i}{\|\mathbf{r}_i - \mathbf{r}_j\|} A_{ij} + HV_i = m_i C_i \frac{d\theta_i}{dt} \quad (11)$$

where K is the thermal conductivity and A_{ij} represents the contact area between particles i and j . If the particles intersect, the contact area is given by

$$A_{ij} = A_0 \frac{R_i + R_j}{\|\mathbf{r}_i - \mathbf{r}_j\|} \quad (12)$$

$$\text{where } A_0 = \frac{\pi(R_i + R_j)^2}{8} \text{ and } A_{ij} \leq A_{max} = \frac{\pi(R_i + R_j)^2}{4}.$$

The laser heating term in Equation 11 is given by

$$HV_i = \alpha I A_i \quad (13)$$

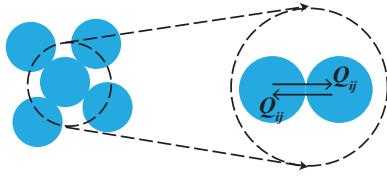


Fig. 3. Particle-to-particle heat conduction

where α is the absorptivity ($0 \leq \alpha \leq 1$), I is the laser intensity in Watts per square meter, and A_i is the incident area of the laser. The laser intensity is given via the Beer-Lambert law

$$I = I_0 e^{-\alpha z} \quad (14)$$

where I_0 is the intensity of the incident light, a is the absorption coefficient (a material property), and z is the penetration depth.

2.3. Thermal Effects – Phase Change

In laser sintering it is important to recognize that often times the powder particles will melt, especially the smaller ones. Thus when solving the energy balance equation (Eq. 11) it is necessary to update the heat capacity, C , as the material changes state. Generally speaking, $C_{\text{gas}} > C_{\text{liquid}} > C_{\text{solid}}$. Also during phase changes, the latent heat of melting, solidification, vaporization, or condensation (depending on the situation) must be added to the heat capacity. The general algorithm used by the authors, and previously used in [15], to deal with phase changes is outlined as follows:

1. *Solid*: If $\theta(t) < \theta_m$ and $\theta(t + \Delta t) < \theta_m$, then $C(\theta) = C_{\text{solid}}$.
2. *Melting*: If $\theta(t) < \theta_m$ and $\theta(t + \Delta t) \geq \theta_m$, then $C(\theta) = C_{\text{solid}} + \frac{\Delta H_m}{\delta\theta}$.
3. *Liquid*: If $\theta(t) \geq \theta_m$ and $\theta(t + \Delta t) < \theta_v$, then $C(\theta) = C_{\text{liquid}}$.
4. *Solidification*: If $\theta(t) \geq \theta_m$ and $\theta(t + \Delta t) < \theta_m$, then $C(\theta) = C_{\text{liquid}} + \frac{\Delta H_s}{\delta\theta}$.
5. *Vaporizing*: If $\theta(t) < \theta_v$ and $\theta(t + \Delta t) \geq \theta_v$, then $C(\theta) = C_{\text{liquid}} + \frac{\Delta H_v}{\delta\theta}$.
6. *Gas*: If $\theta(t) \geq \theta_v$ and $\theta(t + \Delta t) \geq \theta_v$, then $C(\theta) = C_{\text{gas}}$.
7. *Condensation*: If $\theta(t) \geq \theta_v$ and $\theta(t + \Delta t) < \theta_v$, then $C(\theta) = C_{\text{gas}} + \frac{\Delta H_c}{\delta\theta}$.

In the above cases θ_m is the melting temperature, θ_v is the vaporization, or boiling, temperature, ΔH_m is the latent heat of melting, ΔH_s is the latent heat of solidification, ΔH_v is the latent heat of vaporization, and ΔH_c is the latent heat of condensation. The latent heat terms are added to account for

the energy necessary for phase transformations to occur. They act to essentially keep the temperature constant as the material changes phase.

In the case where the temperatures at the present and future time steps are greater than the boiling temperature, the particle is treated as a gas and is assumed to ‘disappear’. In this case that particle no longer interacts with other particles and is removed from the simulation.

2.4. Numerical Solution Scheme

The model presented here is discretized in time and solved via an explicit fourth order Runge-Kutta scheme. There are many references available for the reader interested in more details about this scheme, including [16].

3. Simulation Results and Discussion

The deposition and subsequent laser sintering of a single layer of particles was carried out via the described framework. Deposition is modeled by first randomly placing 4000 particles in a domain. Note that the particles diameters were constrained to be between 20 and 100 microns. The particles were dropped from a height of 0.6 to 0.9 mm on to a flat floor. Gravity causes the particles to fall and distribute themselves in a layer. After deposition, the ensuing layer of particles had a non-uniform thickness of approximately 100 microns. The laser intensity was assumed to be uniform throughout the cross-sectional area of the circular beam and the particles were assumed to have no thermal interaction with the surrounding atmosphere or walls. Table 1 lists more parameters values used in this simulation.

Table 1. Parameter values used in this simulation

Particle Diameter (mean, std. dev.)	0.05 mm, 0.02 mm
Domain Cross-Sectional Dimensions	2 mm x 2 mm
Number of Particles	4000
Initial Powder Temperature	373 K
Laser Power	50 W
Scan Speed	1.5 m/s
Laser Beam Diameter	0.5 mm
Particle Density	7800 kg/m ³
Melting Temperature	1800 K
Vaporization Temperature	3500 K
Thermal Conductivity (solid, liquid)	40 W/m-K, 60 W/m-K
Heat Capacity (solid, liquid)	600 J/kg-K, 900 J/kg-K

Figure 4 shows screenshots of the deposition and sintering of a layer of particles. In the simulation it is apparent that smaller particles heat up much quicker than larger ones and particles near the top of the layer are initially hotter since they receive more energy from the laser. A couple other trends can also be easily noticed from this figure, namely:

- The temperature distribution starts out highly uneven but begins to equilibrate as conduction takes place,
- The time scale for heating due to the laser is much quicker than that due to conduction.

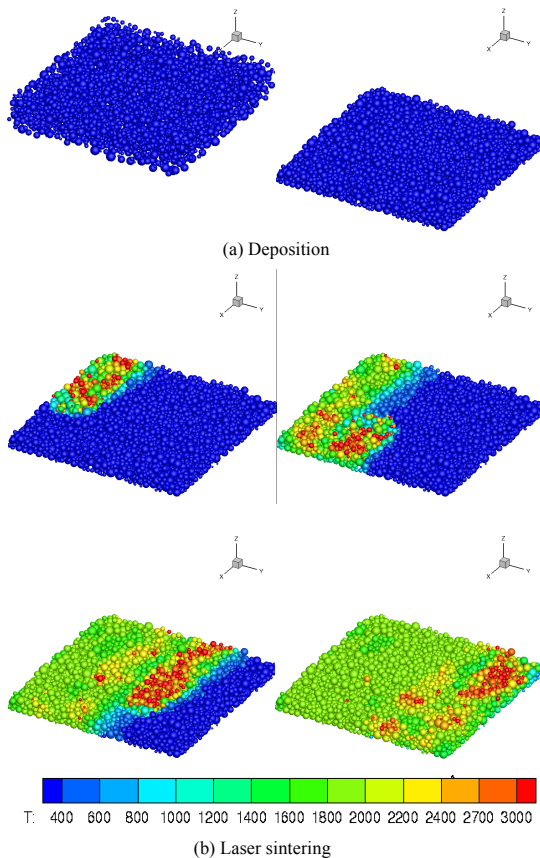


Fig. 4. (a) Deposition (top row) and (b) subsequent laser sintering of a 2 mm x 2 mm box of powdered materials, color represents temperature (scale ranges from 400 to 3000 K)

Note that in this simulation the circular laser beam moves in a zig-zag pattern across the length of the domain. Currently, the laser beam intensity is assumed to be uniform throughout its cross-sectional area; however, in future simulations the intensity of the laser beam can be made to decrease further away from its center point, as one would expect to see in a typical CO₂ laser.

Figure 5 displays the average temperature of the entire particle bed as a function of the simulation time. In this plot it can be seen that the average bed temperature increases linearly as the laser passes over it. Note that the periodic plateaus in Figure 5 occur when the laser passes out of the domain before it reverses direction and re-enters.

Table 2 displays the distribution of particle sizes, the average heights of the particle center (directly related to laser radiation received), and the average temperature of those particles at the end of the simulation.

From this table it can be seen that smaller particles tend to be hotter, though this is complicated by particle height and location, as the temperatures tend to equilibrate as time passes and conduction takes place.

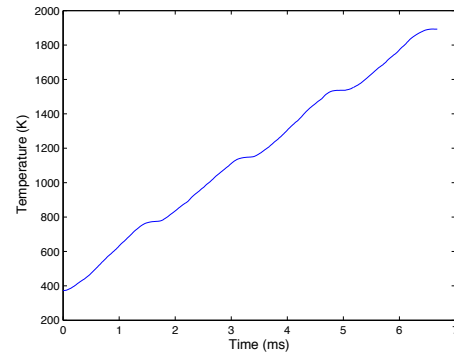


Fig. 5. Average temperature of entire powder bed with time

Table 2. Results at the end of the simulation discretized by particle size

Particle Diameter (μm)	Number of Particles	Avg. Height of Particle Center (μm)	Average Temperature (K)
20 – 30	653	76.4	1949
30 – 40	613	80.3	1849
40 – 50	802	85.2	1905
50 – 60	736	86.4	1891
60 – 70	554	86.4	1894
70 – 80	357	86.5	1873
80 – 90	186	79.9	1854
90 – 100	99	74.5	1835

Since thermal conductivity and heat capacity are bulk material properties, additional simulations were run varying the thermal conductivity and heat capacity of the individual particles. The mean value of these parameters remained the same but the specific values for individual particles were varied. For thermal conductivity a random distribution of $k_{\text{solid}} = 40 \pm 10$ W/m-K was used for each particle. For heat capacity a random distribution of $C_{\text{solid}} = 600 \pm 100$ J/kg-K was used. However, the simulation results showed no appreciable difference in the temperature profile of the final particle bed when these parameter values were varied. This finding supports use of the assumption that all particles of a specific material may be attributed the same thermal conductivity and heat capacity.

4. Conclusions and Extensions

In this work a basic framework has been set up to model the deposition and heat transfer involved in the process of selective laser sintering. A discrete element model is employed where the powdered particles to be sintered are represented by individual spheres of varying sizes. The spheres are dropped into a box to represent the deposition process. Subsequently a laser is passed over them in a zig-zag pattern. The laser intensity is assumed to be uniform and penetration as a function of depth is modeled via the Beer-Lambert Law. Particle-to-particle heat conduction is solved for. It is assumed that the particles are small enough such that there is a uniform temperature distribution within each sphere. Additionally, thermal interaction with the surrounding atmosphere and domain walls are currently not considered.

From this simulation it is easy to see the effect of particle size on its temperature. Smaller particles heat up much quicker and will subsequently melt first. After the laser has passed over a section of particles, conduction will cause the temperature of the particle bed to become more uniform. The laser power and scan rate greatly affects the rate at which the particles heat up.

This framework is general enough for it to be used in the simulation of a number of different materials, including a mixture of different powdered materials. This can aid in the set-up of physical experiments by being able to provide operators with approximate values for the relevant parameters (i.e. laser power, scan speed, etc.) when trying to laser sinter a certain material, or mixture of materials. Additionally, the thermal governing equations and phase change model described in this paper can be used to simulate other laser processing techniques, such as laser ablation for example.

By building upon this basic framework, there still remains many ways to improve the current model. The laser intensity can be made to diminish further away from its center allowing for investigations into optimal scan spacing to be made. Radiative and convective heat transfer can be included as these will play a role in the cooling and solidification of particles after the laser sintering has been completed. Additionally, the authors are currently investigating methods to model the densification rate and resultant porosity of the powder bed as particles are melted and begin to fill the voids present in the initial bed. Methods for computing the residual stresses that will be present in the final part are also being investigated. Once these are completed, parameter studies can be performed to optimize process parameters for a certain application. Finally, and perhaps most importantly, there remains the need for experimental validation of this model.

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